Problem 4

Consider a 4-dimensional hyper-cubic lattice with the lattice spacing Δ. Compute the "lattice propagator" as the sum over all lattice paths connecting the lattice points 0 and x,

$$\sum_{\text{lattice path } 0\to x} e^{-m_0 L(\text{path})},$$

where $L(\text{path})$ is the number of links in the path. The path may go through a given link any number of times.

Consider the limit $\Delta \to 0$. Observe that at fixed m_0 the sum either diverges, or is dominated by the shortest path from 0 to x. Determine how $m_0 = m_0(\Delta)$ should depend on Δ for long paths to remain relevant in the limit $\Delta \to 0$.

Suggestion: follow the steps outlined in Lecture 7, with suitable modifications to take into account the lattice structure. The representation

$$\delta_{n,0} = \int_{-\pi}^{\pi} \frac{\mathrm{d}k}{2\pi}\, e^{ikn}$$

of the Kronecker delta may be useful.

Solution

Write $x = \Delta y$, where $y \in \mathbb{Z}^4$. Let

$$S = \{+e_1, -e_1, +e_2, -e_2, +e_3, -e_3, +e_4, -e_4\}$$

be the set of the eight allowed link directions. A path with n links has $L(\text{path}) = n$ and weight $e^{-m_0 n}$. Thus the sum over paths is

$$G_\Delta(x) = \sum_{n=0}^{\infty} e^{-m_0 n} \sum_{s_1,\ldots,s_n \in S} \prod_{\mu=1}^{4} \delta_{\sum_{k=1}^{n} s_k^\mu,\, y^\mu}. \tag{1}$$

Here s_k^μ is the μth component of the kth link direction, and $\delta_{a,b}$ is the Kronecker delta. The empty path is included for $n = 0$.

Using the Fourier representation of the four Kronecker deltas gives

$$\prod_{\mu=1}^{4} \delta_{\sum_{k=1}^{n} s_k^\mu,\, y^\mu} = \Delta^4 \prod_{\mu=1}^{4} \left[\int_{-\pi/\Delta}^{\pi/\Delta} \frac{\mathrm{d}P_\mu}{2\pi} \right] \exp\left(iP \cdot x - i\Delta P \cdot \sum_{k=1}^{n} s_k \right). \tag{2}$$

The sum over link directions factorizes over the steps:

$$\sum_{s_1,\ldots,s_n\in S} \exp\left(-i\Delta P\cdot\sum_{k=1}^{n} s_k\right) = \left[\Sigma(P)\right]^n,$$

where

$$\Sigma(P) = \sum_{s\in S} e^{-i\Delta P\cdot s} = 2\sum_{\mu=1}^{4}\cos(\Delta P_\mu).$$

The sum over n is a geometric series. Since $\Sigma(0) = 8$, it converges when $8e^{-m_0} < 1$, or $m_0 > \log 8$. For $m_0 < \log 8$ the sum over paths diverges. Summing the geometric series gives the exact lattice expression

$$G_\Delta(x) = \Delta^4 \prod_{\mu=1}^{4}\left[\int_{-\pi/\Delta}^{\pi/\Delta}\frac{\mathrm{d}P_\mu}{2\pi}\right] e^{iP\cdot x}\,\frac{e^{m_0}}{e^{m_0}-\Sigma(P)}. \tag{3}$$

At the critical value $m_0 = \log 8$, the Fourier integrand is singular at $P = 0$; in four dimensions the path sum at fixed lattice endpoints is still convergent, and this case can be obtained as the limit from $m_0 > \log 8$.

In the limit $\Delta \to 0$, with x fixed, expand

$$\Sigma(P) = 2\sum_{\mu=1}^{4}\cos(\Delta P_\mu) \simeq 8 - \Delta^2\sum_{\mu=1}^{4} P_\mu^2.$$

To retain long paths and obtain a finite continuum mass parameter m, tune

$$e^{m_0} - 8 = \Delta^2 m^2, \qquad m_0 = \log 8 + \frac{\Delta^2 m^2}{8} + O(\Delta^4), \tag{4}$$

with $m^2 > 0$ for convergence (and $m = 0$ understood as the critical limit). Then, for fixed $x \neq 0$,

$$\frac{G_\Delta(x)}{8\Delta^2} \longrightarrow \int_{\mathbb{R}^4}\frac{\mathrm{d}^4P}{(2\pi)^4}\frac{e^{iP\cdot x}}{m^2+P^2}. \tag{5}$$

For $r = |x| > 0$, this is the four-dimensional Green function

$$\int_{\mathbb{R}^4}\frac{\mathrm{d}^4P}{(2\pi)^4}\frac{e^{iP\cdot x}}{m^2+P^2} = \begin{cases}\dfrac{mK_1(mr)}{4\pi^2 r}, & m > 0,\\[2ex] \dfrac{1}{4\pi^2 r^2}, & m = 0,\end{cases}$$

where K_1 is a modified Bessel function. At $x = 0$ the normalized continuum Green function has the usual ultraviolet divergence. Thus it is the normalized propagator $G_\Delta/(8\Delta^2)$ that has a nonzero continuum limit; the unnormalized sum tends to zero at fixed $x \neq 0$. Paths spanning a fixed physical distance typically use $n = O(|x|^2/\Delta^2)$ links, much more than the $O(|x|/\Delta)$ links of a shortest path. The tuning in (4) keeps these long paths relevant.